TFB comprehensive audit, repair and handover (21 October 2026)

Window: 2026-10-20 21:20 to 22:30 UTC (00:20 to 01:30 Riyadh, 21 Oct). Base: `97fa991` (HEAD of `main`, also the live Render deploy `dep-dbbn3au7bikc73949a6g`, app 8.14.5). Branch / PR: `claude/vibrant-bell-3ic3vs`, draft PR https://github.com/emadsaberbahbah-coder/tadawul-fast/pull/809 (commit `ae95309`). PDF copy: `docs/reviews/TFB_Comprehensive_Audit_2026-10-21.pdf`. No workbook cell, Render setting, environment variable, deploy, provider or order was touched. IBKR was not used (per the assignment); broker reconciliation is therefore **not verified** in this report.

Status labels: FOUND · PREPARED · APPLIED · TESTED · VERIFIED IN PRODUCTION · BLOCKED · ROLLED BACK. "APPLIED" here means committed to the PR branch; nothing in this change set is in production until the owner merges the PR and deploys (Render auto-deploy is off) and installs the two Apps Script files.

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1. الملخص التنفيذي (العربية)

الحالة الحالية: الخدمة على Render تعمل (`/health` سليم، الإصدار 8.14.5 = آخر commit على `main`)، لكن خط المزامنة اليومي لم ينجح في أي تشغيل مجدول منذ 10 أكتوبر (22 من 22 تشغيلاً فشلت بين 13 و20 أكتوبر). أسباب الفشل اليوم: صفحة Top_10 عالقة على حالة "running..." بلا جدول منذ نحو 16:20 UTC، وانخفاض تغطية الأسماء في Global_Markets وMutual_Funds تحت حد 99%، وصفحات القرار أقدم من بياناتها المصدرية. كما سُجّلت 4 أعطال (SIGSEGV) لعمّال الخادم خلال 7 أيام عند استهلاك ذاكرة قارب 95% من الحد.

ما أُنجز (في PR بانتظار الدمج): - إصلاح خطأ مالي جوهري: أوزان My_Portfolio كانت تحوّل أسعار البنس (GBp) بسعر الجنيه، فتضخّم الوزن 100 مرة (93.07% بدلاً من 11.85% في المثال المختبر). أصبح الحل يكشف العملات الفرعية قبل التحويل، ويُفصح عن أسعار الصرف التقديرية في عمود الحالة. - Apps Script: قارئ سجل التكلفة في Portfolio_Decision كان يحتسب الدفعات المباعة/المغلقة؛ وقيم Top_10 كانت تحوّل البنس بسعر الجنيه وتعرض القيمة بالعملة الأصلية على أنها ريال عند غياب سعر الصرف؛ وأي خطأ في التحديث يترك الصفحة على "running...". الثلاثة أُصلحت مع اختبارات. - المزامنة: فشل قراءة "آخر قيمة سليمة" كان يُسجَّل كأنه "لا توجد قيمة سابقة" (149 صفاً فارغاً في Global_Markets). أصبح يُعاد مرة ويُسمّى الفشل صراحة. - حلقة التعلّم: احتساب الربح/الخسارة كان يتجاهل اتجاه التوصية (توصية بيع ارتفع سعرها كانت تُحتسب ربحاً)، وكانت نسبة شارب بوحدات خاطئة (13.42 بدلاً من 2.54)، ومعدل الإصابة بمقام مختلف، ومعدل الدوران في المُقيِّم الظلي يتجاهل الشراء. كلها أُصلحت.

أهم مسألة لم تُحل: لا يوجد تقييم فعلي لنتائج التوقعات في الإنتاج (النضج يتم فقط بتشغيل يدوي)، والسجلات التي فاتها يوم الاستحقاق تبقى معلّقة للأبد. لذلك لا تتوفر حتى الآن أدلة كافية على دقة التوصيات.

جاهزية التوصيات: غير جاهزة للاستخدام كقرار استثماري. خط المزامنة أحمر، وصفحات القرار متأخرة عن مصادرها، وتسوية المحفظة مع الوسيط غير مُتحقق منها، ونتائج التوقعات غير مُقيّمة. تُستخدم المخرجات كبحث فقط إلى أن تُستوفى شروط القسم 8.

إجراءات مطلوبة من المالك (بالترتيب): (1) إعادة تشغيل `refreshDecisionTop10` ومراجعة سجل التنفيذ لتشغيل 16:20 UTC؛ (2) ضبط `TFB_SEC_USER_AGENT` وتشغيل تجديد أسماء الأدوات قبل 21 أكتوبر 07:44 UTC؛ (3) تحديث My_Portfolio بإزالة المركزين المغلقين (إن لم يُحدَّث بعد)؛ (4) مراجعة الـPR ودمجه ثم النشر وتثبيت ملفي Apps Script.

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2. Baseline (before any change)

2.1 Access

Resource	Status	Evidence
GitHub repo	ACCESSIBLE	clone at 97fa991; Actions runs and logs read via the GitHub connector
Render `tadawul-fast-bridge`	ACCESSIBLE (read)	service, deploys, events, metrics, app logs (workspace id passed per call)
Google Sheet `1QaOSN…r2HwsG04`	PARTIAL	Drive metadata readable (last modified 2026-10-20 20:54 UTC / 23:54 Riyadh); content read returned only _Diag_Response_Shape; xlsx/csv export "not implemented"; no Sheets API credential in this environment
Apps Script project `1IvEj8…2-`	BLOCKED	Drive: "Requested entity was not found" (bound project not reachable); repo `apps_script/` used instead
Dashboard `tfb-investment-platform…chatgpt.site`	BLOCKED	HTTP 401 "Sign in required" (ChatGPT login); front-end source not in this repo
IBKR	NOT USED	per assignment

2.2 Architecture as verified

Providers (EODHD, Yahoo chart/fundamentals, others) → FastAPI service on Render (`main.py`, `core/data_engine_v2.py`, `routes/*`) → GitHub Actions `daily_sync.yml` (`scripts/run_dashboard_sync.py`) writes the market pages and My_Portfolio → Apps Script producers (`10_My_Portfolio.gs` → Portfolio_Decision, `16_Decision_Top10.gs` → Top_10 cockpit, `26_Decision_Chain.gs`, `24_Sync_Dispatch.gs`, `11_Manual_Refresh_Coordinator.gs`) → audits (`audit_full_refresh_coverage.py`, `audit_decision_surface_freshness.py`, `finalize_decision_publication.py`, `validate_dashboard.py`) → the dashboard reads the Sheet through the restricted relay (`routes/platform_source.py`, `integrations/platform_source_read.py`).

Authoritative sources: holdings and cost basis = My_Portfolio + `_Portfolio_CostBasis` ledger (owner data); SAR money on the decision surface = `core/analysis/portfolio_actions.py` with FX from `opportunity_builder._resolve_fx`, gated by `core/portfolio_reconciliation.certify_portfolio_inputs`; My_Portfolio Value/Cost/P&L stay in native currency, with SAR-converted weights from `data_engine_v2._compute_portfolio_fields`; Top_10 SAR holdings from `16_Decision_Top10.gs`.

2.3 Baseline metrics (denominators and periods stated)

Metric	Value	Source / period
Dashboard Advanced Sync, scheduled runs	**0 / 22 succeeded**	13 Oct 17:17 to 20 Oct 17:17 UTC; last success 10 Oct 10:40 UTC (run 38045703392)
Provider Target Coverage Alert	0 / 5 succeeded	16 to 20 Oct
Render `/health`	200, healthy, 0.56 s	20 Oct 21:2x UTC
HTTP status mix	200: 1,563 (97.6%); 499: 29; 502: 6 (all before the 13:52 deploy); 401: 4	24 h to 20 Oct 21:30 UTC, n = 1,602
HTTP latency p50/p95/p99	**not available** (Render returns empty latency series; no request logs)	—

Metric	Value	Source / period
Worker crashes (gunicorn code 139)	4	7 days; memory at 95% of the 4.29 GB limit before the 20 Oct 09:30 crash; 87% at 21:25 UTC
identity_guard refusals	Mutual_Funds 71, Global_Markets 14, Commodities_FX 13	18 Oct 16:03 to 20 Oct 17:47
Name coverage (floor 99%)	Mutual_Funds 97.90%, Global_Markets 97.84%	evening sync, 20 Oct
Decision surface	Top_10 status RUNNING (since ~16:20 UTC), PF 10:40 Riyadh vs My_Portfolio 20:19 Riyadh	Decision Surface Freshness run 39474537234
Test suite	**5,355 passed, 36 skipped, 0 failed**	local, pinned requirements, Python 3.11
Open PRs	30 (4 drafts), oldest 6 Oct	GitHub

3. Findings by layer

3.1 Scheduled workflows (FOUND)

1. **Top_10 cockpit stuck RUNNING** — the run started ~16:20 UTC never finished; the page holds 16 rows and no grid. `validate_dashboard.py:985-1012` cannot find the `Rank|Symbol` header, and because every sync leg validates Top_10, all three legs of run 39464797610 went red although each sync wrote cleanly (Global_Markets 6,613 rows, exit 0). Root cause in code: `refreshDecisionTop10()` wrote `running…` first and had no failure path. → fixed in `16_Decision_Top10.gs` v1.13.10 (§6, item 3). The 16:20 run itself needs an owner re-run.
2. **Global_Markets name floor (new)** — `[ID-FIREWALL]` quarantined 149 rows and `[FW-KEEP]` restored 0/149 "no last-good"; the morning run restored 134/135. The same leg logged `OHLC-READBACK … UNAVAILABLE (read_failed)`. Root cause in code: `SheetsWriter.read_values` returns `None` on any API error and `_keep_last_good_rows` treated it as an empty sheet. → `run_dashboard_sync.py` v6.70.1 (§6, item 4).
3. **Mutual_Funds name floor** — the reviewed captions expired 19 Oct 12:31 UTC (known; 16 Global_Markets captions also expired then, 67 more expire **21 Oct 07:44 UTC**). Owner action: `TFB_SEC_USER_AGENT` + renewal dispatch.
4. **Decision freshness** — `PF_RUN_STALE`, `PF/T10_OLDER_THAN_SOURCE`, `T10_POOL_RECONCILIATION_UNPROVEN`, consumption receipts unproven. Owner action: re-run both producers after the sync; install `26_Decision_Chain`.
5. **Provider Target Coverage** — deadlock: the workflow only saves a baseline on exit 0 and cannot exit 0 without one (`baseline_at_utc 2026-08-18`). Owner action: dispatch with `bootstrap=true` after a healthy sync.

3.2 Render (FOUND)

- Healthy startup at 13:52 UTC (7 route modules, 110 signatures, no startup warnings). CPU peak 0.80 of 2 cores.
- **Memory is the operational risk**: 4 worker SIGSEGV (code 139) in 7 days; a single `POST /sheet-rows/opportunity-candidates` at 20:24 UTC added ~1.39 GB that was not released. Not a Render `server_failed` event, so it is invisible in the event feed. Not fixed here (needs profiling; see backlog B2).
- Commodities_FX: Yahoo returns 404 for `GRT-USD`, `COMP-USD`, `UNI-USD`, `SUI-USD`, `IMX-USD`, `STX-USD`, `APT-USD`. Verified on 20 Oct that Yahoo serves them as `GRT6719-USD`, `COMP5692-USD`, `UNI7083-USD`, `SUI20947-USD`, `IMX10603-USD`, `STX4847-USD`, `APT21794-USD` (names match). A fixture shows bare `UNI-USD` once resolved to an unrelated "UNICORN Token". The project already carries a native EODHD crypto route (`TFB_EODHD_CRYPTO_MAP`, default off); not changed here (backlog B6).
- Mutual_Funds refusals are stale-universe rows (matured iBonds, renamed ETFs: IBHA, IBDO, IBMJ, IBTD, KLD, IGN, SPLG…) — a universe problem, not a guard defect (backlog B7).

3.3 Financial logic (FOUND → APPLIED where marked)

#	Defect	Severity	Status
F1	`data_engine_v2._compute_portfolio_fields` upper-cased GBp to GBP before the FX lookup: pence valued at the pounds rate, weight 100x (93.07% vs 11.85%), drives REDUCE/ADD	Critical	APPLIED, TESTED
F2	`16_Decision_Top10.gs dt10HoldingFromRow_`: same subunit bug, and a missing FX rate kept the native amount labelled `value_sar` (HKD 10,000 → "10,000 SAR") feeding budget base and sector caps	Critical	APPLIED, TESTED (needs install)
F3	`10_My_Portfolio.gs _myPortfolioLedgerMap_` counted Sold/Closed lots (qty 20 / cost 90.5 instead of 10 / 100.5)	High	APPLIED, TESTED (needs install)
F4	`portfolio_fx_status` read `OK:SAR` even when built-in INDICATIVE (floating) rates were used	Medium	APPLIED, TESTED
F5	Missing price + carried stale `position_value` still drives weights / REDUCE (`TFB_MP_BLOCKED_NULLS` default off)	High	FOUND (backlog B4)
F6	P&L % unit contract disagrees: schema says fraction, engine and sync write percent points	Medium	FOUND (backlog B5)
F7	Two built-in FX tables disagree (GBP 4.80 vs 4.75, EUR 4.05 vs 4.10)	Medium	FOUND (backlog B5)
F8	Weight denominators differ by design (My_Portfolio holdings-only, Portfolio_Decision holdings+cash)	Low (definition)	FOUND, documented here
F9	Realized P&L, income/dividends, fees and FX effects are not separated anywhere; cost is converted at today's FX, so SAR P&L includes currency moves	High (reporting gap)	FOUND (backlog B8)

Consolidated SAR today: the certified Portfolio_Decision path is subunit-safe and fails closed without a fresh evidence packet; it is the only SAR money surface and is BLOCKED by default without broker evidence (`PD_IBKR_EVIDENCE` off). Per-share entry/stop/target stay in the trading currency.

3.4 Recommendations and learning loop (FOUND → APPLIED where marked)

#	Defect	Status
L1	Maturation runs only on a manual dispatch with `run_audit=true`; a record whose target session was missed can never mature (only `price_unavailable` expires)	FOUND (backlog B1, highest learning priority)
L2	Win/loss ignored the call's direction (two SELLs that rose counted 100% wins)	APPLIED, TESTED (`track_performance` v6.46.0)
L3	Entry-sanity check vs the median of all prior entries (±35%) locks a symbol out after any large move or bonus issue → survivorship	FOUND (backlog B3)
L4	Sharpe/Sortino treated 7-90-day percent returns as daily fractions (13.42 vs 2.54)	APPLIED, TESTED
L5	Hit rate by horizon included breakevens in the denominator (25 vs 50)	APPLIED, TESTED
L6	Top_10 page fills to 10 from criteria-failing pools (labelled `T1/T2 BACKFILL[…]` in the selection reason; the tracker does not persist the tier)	FOUND (backlog B9)
L7	Shadow-scorer turnover/cost counted exits only (adding a name cost 0)	APPLIED, TESTED (`run_shadow_scorer` v1.9.4)
L8	Scorer's "last daily close" can be an intraday bar (cron 15:20 UTC)	FOUND (backlog B10)

#	Defect	Status
L9	Matured duplicates counted twice by default (TRACK_DEDUP_MATURED off)	FOUND (backlog B10)
—	Tracker: no fees/slippage, no benchmark, no model/rule version on records; horizons are calendar days vs policy trading days	FOUND

Forecast performance: **not reported**. Matured observations are produced only by manual audits and include the direction and duplicate defects above, so any accuracy figure from the current log would be misleading.

3.5 Dashboard and Sheet (page-by-page)

The dashboard is BLOCKED (login), so a visual page-by-page assessment was not possible. From the relay code (`integrations/platform_source_read.py READ_LIMITS`): the dashboard can read `_Status`, `Top_10_Investments`, `Portfolio_Decision`, `My_Portfolio`, `_Portfolio_CostBasis`, `_Cash_Snapshot`, `Daily_Review_2026_10_14`, plus up to 20 rows of `Market_Leaders`/`Global_Markets`.

- `Portfolio_Decision_v2` / `Top_10_Investments_v2` are **not readable** by the relay (FOUND, backlog B11).
- The daily-review tab is pinned to `Daily_Review_2026_10_14` (FOUND, backlog B11).
- `_Diag_Response_Shape` reports `INCOMPLETE-COLUMNS` for Global_Markets (5 concerning blank columns).
- Whether My_Portfolio still lists the two US positions closed on 19 Oct could not be checked (Sheet content not readable here). `docs/apps-script-source-prerequisites.md` is stale (lists `10_My_Portfolio.gs` as unavailable).

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4. Implementation register

#	Root cause	Change	Files	Tests	Status
1	F1, F4	resolve subunit token case-sensitively before FX (`_pf_subunit_major`), divide by 100; unresolvable subunit falls into the existing whole-book `PORTFOLIO_FX_INCOMPLETE` fail-closed path; status names INDICATIVE rates	`core/data_engine_v2.py` 5.154.0 → **5.155.0**	`tests/test_pf_weight_subunit_fx.py` 6 passed; 4 of 6 fail on the old engine (golden negative)	APPLIED, TESTED
2	F3	ledger reader excludes `inactive/closed/sold` (blank = active, as `ledger_symbols`)	`apps_script/10_My_Portfolio.gs` 1.5.13 → **1.5.14**	`tests/js/gas_ledger_fx_failed_state.test.js` 10/10; `tests/test_gas_ledger_fx_failed_state.py` 3 passed	APPLIED, TESTED; install = owner
3	F2 + stuck RUNNING	`dt10HoldingFxRate_` (subunit-safe, mirrors `_resolve_fx`); missing rate → `value_sar: null` + `_input_incomplete`; `refreshDecisionTop10` wraps the body and writes a terminal `status: FAILED | failed_at_utc=… | <error>`, `_Status` ERROR, clears the run marker, rethrows (all three steps individually guarded). A hard 6-minute Apps Script timeout cannot be caught.	`apps_script/16_Decision_Top10.gs` 1.13.9 → **1.13.10**; JS fingerprints/counts in `tests/js/*`	same JS suite; all 10 JS suites green	APPLIED, TESTED; install = owner
4	last-good read failure misreported	one retry (3 s) on `read_values` = None; `KLG-READ` warning; FW-KEEP says "last-good READ FAILED"	`scripts/run_dashboard_sync.py` 6.70.0 → **6.70.1**	`tests/test_sync_klg_read_retry.py` 3 passed	APPLIED, TESTED
5	L2, L4, L5	direction-aware outcome (`_directional_outcome`; HOLD → `neutral_records`), per-horizon Sharpe/Sortino in fraction units, hit rate = W/(W+L). The Performance_Log `Outcome` column and ROI values are **unchanged**	`scripts/track_performance.py` 6.45.0 → **6.46.0**	`tests/test_learning_metric_fixes.py` 37 passed	APPLIED, TESTED
6	L7	one-way turnover = Σ	Δw	/2; cost charged on buys and sells (full swap still 0.36%)	`scripts/run_shadow_scorer.py` 1.9.3 → **1** `tests/test_s1_*`, `tests/test_shadow_scorer_regret_read_`
7	CI	enrol the new suites in the blocking lanes	`.github/workflows/ci.yml` 1.1.27 → **1.1.28**; `scripts/verify_deployment.py` pins	—	APPLIED

Rollback: every change is a version bump on one file with its predecessor in `97fa991`; revert the PR commit (or re-install `10_My_Portfolio.gs` v1.5.13 / `16_Decision_Top10.gs` v1.13.9). No data migration, no new flag, no new dependency, no cost.

Behaviour boundaries to note in the register: from the deploy day, (a) Shadow_History cost drag charges buys (historical rows untouched), (b) tracker win rates and calibration buckets score SELL-family calls by direction and exclude HOLD, (c) My_Portfolio Actual Weight % of `.L`/JSE/TASE holdings drops by 100x to the correct value.

Production verification: **not possible in this run** — the PR is not merged and Render auto-deploy is off. After deploy, verify: `/health` app/engine versions; one My_Portfolio refresh showing `portfolio_fx_status`; the next sync log containing `KLG-READ` only on a real read failure; one manual `refreshDecisionTop10`

ending in OK.

Final verification (integrated tree)

- Whole suite: **5,404 passed, 36 skipped, 0 failed** (baseline at `97fa991`: 5,355 passed, 36 skipped, 0 failed; +49 new tests).
- All JS suites under `tests/js/` and `tests/test_dt10_calendar_completeness.js` pass; `apps_script_syntax` compiles the 5 sources.
- `scripts/audit_repository_workflows.py --fail-on-warnings`: same 2 pre-existing `daily_sync.yml` warnings as `main`.

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5. Before / after

Measure	Before	After (code)	Production
VOD.L weight in the F1 example	93.07%	11.85%	after deploy
Ledger qty/cost in the F3 example	20 / 90.5	10 / 100.5	after install
HKD holding without a rate on Top_10	10,000 "SAR"	null + funding withheld	after install
Top_10 status after a thrown error	RUNNING forever	FAILED with UTC time and error	after install
Two SELLs that rose +5%/+3%	100% win rate	0%	next tracker audit
1M Sharpe on [2,-1,3,1]%	13.42	2.54	next tracker report
Basket {A} → {A,B} turnover / cost	0 / 0	50% / 0.18%	next scorer run
Sync success rate	0 / 22	unchanged until the owner actions in §7 are done	—

No overall accuracy or improvement percentage is claimed.

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6. Prioritized remaining backlog

ID	Item	Impact	Depends on	Acceptance criteria	Effort
B1	Schedule the tracker audit; mature a missed target session from the historical daily close (or EXPIRE with a reason)	Critical for learning	none	scheduled audit runs daily; no ACTIVE record older than target + 5 sessions	M
B2	Profile `/sheet-rows/opportunity-candidates` memory; as a stop-gap set `TFB_MAX_REQUESTS/_JITTER` on Render	High (crashes)	owner for env	7 days without code 139; memory < 80%	M
B3	Entry-sanity check against a rolling window / independent quote / corporate-action adjustment	High (survivorship)	none	a 40% move does not exclude future picks	S
B4	Null price-derived fields and suppress book weights when a held price is missing (arm `TFB_MP_BLOCKED_NULLS` after a dry run)	High	none	missing price never produces a weight or REDUCE	S
B5	One P&L % unit across schema/engine/sync/audit; one shared FX table with source and as-of on the sheet	Medium	owner decision	audit_workbook no longer flags valid rows; FX source/time visible	M
B6	Crypto identity on Commodities_FX: arm `TFB_EODHD_CRYPTO_MAP` after a dry run, or add the verified Yahoo IDs	Medium	dry run	0 quarantines for the 7 tokens	S
B7	Retire matured/renamed Mutual_Funds symbols via `universe_maintenance`	Medium	owner review	no `pre_existing_blank_shell` refusals	S
B8	Separate realized P&L, income, fees, FX effect and external flows; historical-FX cost basis	High (reporting)	broker statements	reconciled SAR bridge per position	L
B9	Persist Top_10 tier/backfill flag on tracker records; publish "No qualifying opportunity" when no row passes	Medium	none	backfill outcomes separable	S
B10	Completed-session check in the scorer's `_last_bar`; arm `TRACK_DEDUP_MATURED`	Medium	none	no intraday close; duplicates counted once	S
B11	Relay: v2 decision tabs and a non-dated daily-review tab (security review first)	Medium	owner	dashboard shows v2 pages	S
B12	Relabel the Performance_Log `Outcome` column by direction behind a gate (calibration_report, tfb_backtest read it)	Medium	B1	backtests score SELL calls by direction	M
B13	Fees/slippage and a TASI/SPUS benchmark in the tracker; trading-day horizons per policy §7.1	Medium	B1	net-of-cost results vs benchmark with CIs	M
B14	Triage the 30 open PRs (oldest 6 Oct); several overlap this area	Medium	owner	≤ 10 open PRs with a decision each	M

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7. Owner actions (cannot be done from this session)

1. Apps Script Executions: inspect the ~16:20 UTC `refreshDecisionTop10` failure; re-run Top_10 and Portfolio_Decision.
2. Before **21 Oct 07:44 UTC**: set repository variable `TFB_SEC_USER_AGENT`, dispatch `Verified Name Label Renewal` with `mode=renew`, commit the renewed registry.
3. My_Portfolio: confirm the two US positions closed on 19 Oct are removed.
4. Review and merge the PR; deploy on Render; install `10_My_Portfolio.gs` v1.5.14 and `16_Decision_Top10.gs` v1.13.10.
5. Dispatch Provider Target Coverage with `bootstrap=true` after the first healthy sync.

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8. Final verdict

- **Operationally healthy: No.** The API is up, but the scheduled sync has not succeeded since 10 Oct, the decision cockpit is stuck, and workers crash under memory pressure.
- **Financially consistent: Not yet.** Three confirmed currency/ledger defects are fixed in the PR but not deployed; P&L components are not separated; broker reconciliation was not verified.
- **Validated for recommendation use: No.** Forecast outcomes are not evaluated in production, and until B1 is done there is no sample from which to judge accuracy. Treat Top_10 and Portfolio_Decision as research output only.

Readiness criteria: a green scheduled sync on 3 consecutive days; the PR deployed and installed; B1 running with at least 30 matured, direction-scored, de-duplicated records per horizon; Portfolio_Decision certified against a current broker statement.